MATH 152 **QUIZ 5** NAME _____

Mrs. Bonny Tighe 25 points

7.7 TA_____ Wed 10/12/2022

1. Use (a) the Midpoint Rule (b) the Trapezoidal Rule and (c) Simpson's Rule to approximate the given integral with the specified value of n. Find the error bounds for all three methods. $\int_1^2 \sqrt{x}\,dx, \quad n = 4$

Suppose $|f''(x)| \le K \quad for \quad a \le x \le b$, then $|E_T| \le \dfrac{K(b-a)^3}{12n^2}$ $\quad |E_M| \le \dfrac{K(b-a)^3}{24n^2}$

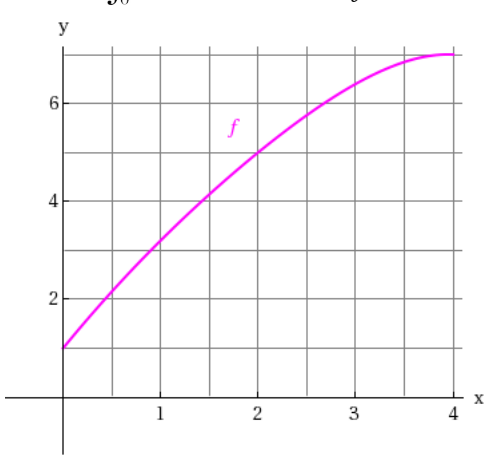
Suppose $|f''''(x)| \le K \quad for \quad a \le x \le b$ then $|E_S| \le \dfrac{K(b-a)^5}{180n^4}$

2. How large should we take n in order to guarantee that the Midpoint approximation for $\int_1^2 (1/x)dx$ is accurate to within 0.001?

8. A radar gun was used to record the speed of a runner during the first 5 seconds of a race (see the table). Use Simpson's Rule to estimate the distance the runner covered during those 5 seconds.

| $t$ (s) | $v$ (m/s) | $t$ (s) | $v$ (m/s) |
|---|---|---|---|
| 0 | 0 | 3.0 | 10.55 |
| 0.5 | 4.65 | 3.5 | 10.69 |
| 1.0 | 7.36 | 4.0 | 10.72 |
| 1.5 | 8.88 | 4.5 | 10.83 |
| 2.0 | 9.79 | 5.0 | 10.83 |
| 2.5 | 10.24 | | |

3. $I = \int_0^4 f(x)dx$ where $f$ is the function whose graph is shown.




*(a)* Use the graph to approximate $M_4$, $L_4$, $R_4$

(b) Which of these are underestimates or overestimates of $I$?

MATH 152 **QUIZ 5A** NAME _____

Mrs. Bonny Tighe 25 points

7.7 TA_____ Wed 10/12/2022

1. Use (a) the Midpoint Rule (b) the Trapezoidal Rule and (c) Simpson's Rule to approximate the given integral with the specified value of n. Find the error bounds for all three methods. $\int_2^3 \sqrt{x}\,dx, \quad n=4$

Suppose $|f''(x)| \le K \text{ for } a \le x \le b$, then $|E_T| \le \frac{K(b-a)^3}{12n^2}$ $|E_M| \le \frac{K(b-a)^3}{24n^2}$

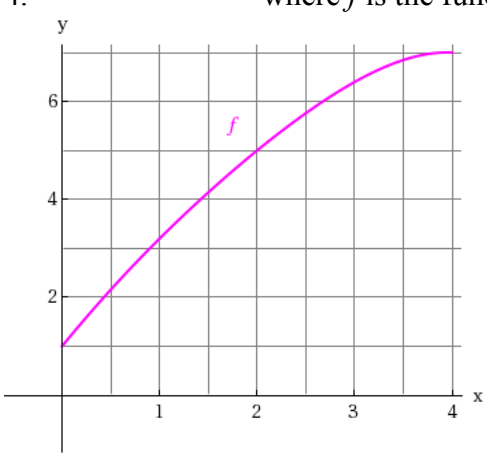
Suppose $|f''''(x)| \le K \text{ for } a \le x \le b$ then $|E_S| \le \frac{K(b-a)^5}{180n^4}$

2. How large should we take n in order to guarantee that the Midpoint approximation for $\int_1^2 \left(\frac{1}{x^2}\right) dx$ is accurate to within 0.001?

3. The table shows values of a force function $f(x)$, where $x$ is measured in meters and $f(x)$ in newtons. Use Simpson's Rule to estimate the work done by the force in moving an object a distance of 18 m.

| $x$ | 0 | 3 | 6 | 9 | 12 | 15 | 18 |
|---|---|---|---|---|---|---|---|
| $f(x)$ | 9.6 | 9.4 | 8.7 | 8.1 | 7.8 | 7.5 | 7.4 |

4. $I = \int_0^4 f(x)\,dx$ where $f$ is the function whose graph is shown.




(a) Use the graph to approximate $M_4$, $L_4$, $R_4$

(b) Which of these are underestimates or overestimates of $I$?